

How to define Null Hypothesis

Null Hypothesis

Default state, statement no change, original Value, original status

Alternate Hypothesis:

Statement which has changed, new status

Solar Example :

Himachal

Company claim avg 100 energy produce per day.

Customer came to complained avg 80 produces per day.

Null Hypothesis

Produces avg 100 energy produces per day.

Alternate Hypothesis

Avg 80 produces per day

Sample mean < population mean

Sample mean > population mean

Right, left, two tailed.

Left

80 < 100

Solar Example :

Rajasthan

Company claim avg 100 energy produce per day.

Customer came gave feedback avg 120 produces per day.

Null

Produces 100 energy

Alternate

Produces 120

Right Tail

Sample mean > population mean

120 > 100

Why do we have to do this?

Region wise stragegy can be changed.

Himachal 80

Rajasthan 120

Left

Null hypothesis: $H_0: \mu = 1000$ Hrs . claim of the company

We can say that Bulb is performing as per standard which is avg 1000Hrs.

Alternative hypothesis: $H_1: \mu < 1000$ (sample mean = 980 Hrs)

Bulb is not performing as per standard (avg is less than 1000Hrs)

p-value < significance level Null Hypothesis is rejected.

p-value > significance level then Null Hypothesis is Accepted

Test

0.0228 < 0.050

0.1111 > 0.050